Chapter 17

Application of Derivatives

Only One Option Correct Type Questions

1. The tangent to the curve $y = e^x$ drawn at the point (c, e^c) intersects the line joining the points $(c - 1, e^{c-1})$ and $(c + 1, e^{c+1})$ **[IIT-JEE-2007 (Paper-1)]**

   (A) On the left of $x = c$
   (B) On the right of $x = c$
   (C) At no point
   (D) At all points

2. The total number of local maxima and local minima of the function

   $$f(x)=\begin{cases}(2+x)^3, & -3<x\le -1\\ x^{\frac{2}{3}}, & -1<x<2\end{cases}$$ is **[IIT-JEE-2008 (Paper-1)]**

   (A) 0
   (B) 1
   (C) 2
   (D) 3

3. Let the function $g:(-\infty,\infty)\to\left(-\frac{\pi}{2},\frac{\pi}{2}\right)$ be given by $g(u)=2\tan^{-1}(e^u)-\frac{\pi}{2}$. Then, g is

   [IIT-JEE)-2008 (Paper-2)]

   (A) Even and is strictly increasing in $(0, \infty)$
   (B) Odd and is strictly decreasing in $(-\infty, \infty)$
   (C) Odd and is strictly increasing in $(-\infty, \infty)$
   (D) Neither even nor odd, but is strictly increasing in $(-\infty, \infty)$

4. Let *f*, *g* and *h* be real-valued functions defined on the interval [0, 1] by $f(x)=e^{x^2}+e^{-x^2}$, $g(x)=xe^{x^2}+e^{-x^2}$ and $h(x)=x^2e^{x^2}+e^{-x^2}$. If *a*, *b* and *c* denote, respectively, the absolute maximum of *f*, *g* and *h* on [0, 1], then **[IIT-JEE-2010 (Paper-1)]**

   (A) $a = b$ and $c \neq b$
   (B) $a = c$ and $a \neq b$
   (C) $a \neq b$ and $c \neq b$
   (D) $a = b = c$

5. The number of points in $(-\infty, \infty)$, for which $x^2 - x\sin x - \cos x = 0$, is **[JEE (Adv)-2013 (Paper-1)]**

   (A) 6
   (B) 4
   (C) 2
   (D) 0

6. Let $f:\left[\frac{1}{2},1\right]\to\mathbb{R}$ (the set of all real numbers) be a positive, non-constant and differentiable function such that $f'(x) < 2f(x)$ and $f\left(\frac{1}{2}\right)=1$. Then the value of $\int_{1/2}^{1} f(x)\,dx$ lies in the interval **[JEE (Adv)-2013 (Paper-1)]**

   (A) $(2e - 1, 2e)$
   (B) $(e - 1, 2e - 1)$
   (C) $\left(\frac{e-1}{2}, e-1\right)$
   (D) $\left(0, \frac{e-1}{2}\right)$

7. The least value of $\alpha \in \mathbb{R}$ for which $4\alpha x^2 + \frac{1}{x} \geq 1$, for all $x > 0$, is **[JEE (Adv)-2016 (Paper-1)]**

(A) $\frac{1}{64}$

(B) $\frac{1}{32}$

(C) $\frac{1}{27}$

(D) $\frac{1}{25}$

8. If $f : \mathbb{R} \to \mathbb{R}$ is a twice differentiable function such that $f''(x) > 0$ for all $x \in \mathbb{R}$, and $f\left(\frac{1}{2}\right) = \frac{1}{2}, f(1) = 1$, then **[JEE (Adv)-2017 (Paper-2)]**

(A) $\frac{1}{2} < f'(1) \leq 1$

(B) $0 < f'(1) \leq \frac{1}{2}$

(C) $f'(1) \leq 0$

(D) $f'(1) > 1$

9. Consider all rectangles lying in the region

$$\left\{(x, y) \in \mathbb{R} \times \mathbb{R} : 0 \leq x \leq \frac{\pi}{2} \text{ and } 0 \leq y \leq 2\sin(2x)\right\}$$

and having one side on the x-axis. The area of the rectangle which has the maximum perimeter among all such rectangles, is **[JEE (Adv)-2020 (Paper-1)]**

(A) $\frac{3\pi}{2}$

(B) π

(C) $\frac{\pi}{2\sqrt{3}}$

(D) $\frac{\pi\sqrt{3}}{2}$

One or More Option(s) Correct Type Questions

10. For the function $f(x) = x\cos\frac{1}{x}, x \geq 1$, **[IIT-JEE-2009 (Paper-2)]**

(A) For at least one x in the interval $[1, \infty)$, $f(x + 2) - f(x) < 2$

(B) $\lim_{x \to \infty} f'(x) = 1$

(C) For all x in the interval $[1, \infty)$, $f(x + 2) - f(x) > 2$

(D) $f'(x)$ is strictly decreasing in the interval $[1, \infty)$

11. If $f(x) = \int_0^x e^{t^2}(t-2)(t-3)dt$ for all $x \in (0, \infty)$, then **[IIT-JEE-2012 (Paper-2)]**

(A) f has a local maximum at $x = 2$

(B) f is decreasing on $(2, 3)$

(C) There exists some $c \in (0, \infty)$ such that $f''(c) = 0$

(D) f has a local minimum at $x = 3$

12. Let $f(x) = x \sin \pi x$, $x > 0$. Then for all natural numbers n, $f'(x)$ vanishes at **[JEE (Adv)-2013 (Paper-1)]**

(A) A unique point in the interval $\left(n, n + \frac{1}{2}\right)$

(B) A unique point in the interval $\left(n + \frac{1}{2}, n + 1\right)$

(C) A unique point in the interval $(n, n + 1)$

(D) Two points in the interval $(n, n + 1)$

13. A rectangular sheet of fixed perimeter with sides having their lengths in the ratio 8 : 15 is converted into an open rectangular box by folding after removing squares of equal area from all four corners. If the total area of removed squares is 100, the resulting box has maximum volume. Then the lengths of the sides of the rectangular sheet are **[JEE (Adv)-2013 (Paper-1)]**

(A) 24 (B) 32

(C) 45 (D) 60

14. The function $f(x) = 2|x| + |x + 2| - |\,|x + 2| - 2|x|\,|$ has a local minimum or a local maximum at x = **[JEE (Adv)-2013 (Paper-2)]**

(A) -2 (B) $\frac{-2}{3}$

(C) 2 (D) $\frac{2}{3}$

15. For every pair of continuous functions $f, g : [0, 1] \to \mathbb{R}$ such that $\max\{f(x) : x \in [0, 1]\} = \max\{g(x) : x \in [0, 1]\}$, the correct statement(s) is(are) **[JEE (Adv)-2014 (Paper-1)]**

(A) $(f(c))^2 + 3f(c) = (g(c))^2 + 3g(c)$ for some $c \in [0, 1]$

(B) $(f(c))^2 + f(c) = (g(c))^2 + 3g(c)$ for some $c \in [0, 1]$

(C) $(f(c))^2 + 3f(c) = (g(c))^2 + g(c)$ for some $c \in [0, 1]$

(D) $(f(c))^2 = (g(c))^2$ for some $c \in [0, 1]$

16. Let $f : (0, \infty) \to \mathbb{R}$ be given by $f(x) = \int_{\frac{1}{x}}^{x} e^{-\left(t+\frac{1}{t}\right)} \frac{dt}{t}$. Then **[JEE (Adv)-2014 (Paper-1)]**

(A) $f(x)$ is monotonically increasing on $[1, \infty)$ (B) $f(x)$ is monotonically decreasing on $(0, 1)$

(C) $f(x) + f\left(\frac{1}{x}\right) = 0$, for all $x \in (0, \infty)$ (D) $f(2^x)$ is an odd function of x on $\mathbb{R}$

17. Let $a \in \mathbb{R}$ and let $f : \mathbb{R} \to \mathbb{R}$ be given by $f(x) = x^5 - 5x + a$. Then **[JEE (Adv)-2014 (Paper-1)]**

(A) $f(x)$ has three real roots if $a > 4$

(B) $f(x)$ has only one real root if $a > 4$

(C) $f(x)$ has three real roots if $a < -4$

(D) $f(x)$ has three real roots if $-4 < a < 4$

18. Let $f, g : [-1, 2] \to \mathbb{R}$ be continuous functions which are twice differentiable on the interval (–1, 2). Let the values of f and g at the points –1, 0 and 2 be as given in the following table:

	$x = -1$	$x = 0$	$x = 2$
$f(x)$	3	6	0
$g(x)$	0	1	–1

In each of the intervals (–1, 0) and (0, 2) the function $(f - 3g)''$ never vanishes. Then the correct statement(s) is(are) **[JEE (Adv)-2015 (Paper-2)]**

(A) $f'(x) - 3g'(x) = 0$ has exactly three solutions in $(-1, 0) \cup (0, 2)$

(B) $f'(x) - 3g'(x) = 0$ has exactly one solution in $(-1, 0)$

(C) $f'(x) - 3g'(x) = 0$ has exactly one solution in $(0, 2)$

(D) $f'(x) - 3g'(x) = 0$ has exactly two solutions in $(-1, 0)$ and exactly two solutions in $(0, 2)$

19. Let $f:\mathbb{R}\to(0,\infty)$ and $g:\mathbb{R}\to\mathbb{R}$ be twice differentiable functions such that f'' and g'' are continuous functions on $\mathbb{R}$. Suppose $f'(2)=g(2)=0$, $f''(2)\neq 0$ and $g'(2)\neq 0$. If $\lim\limits_{x\to 2}\dfrac{f(x)g(x)}{f'(x)g'(x)}=1$, then

[JEE (Adv)-2016 (Paper-2)]

(A) f has a local minimum at $x = 2$

(B) f has a local maximum at $x = 2$

(C) $f''(2)>f(2)$

(D) $f(x)-f''(x)=0$ for at least one $x\in\mathbb{R}$

20. Let $f(x)=\lim\limits_{n\to\infty}\left(\dfrac{n^n(x+n)(x+\frac{n}{2})...(x+\frac{n}{n})}{n!(x^2+n^2)(x^2+\frac{n^2}{4})...(x^2+\frac{n^2}{n^2})}\right)^{\frac{x}{n}}$, for all $x > 0$. Then **[JEE (Adv)-2016 (Paper-2)]**

(A) $f\left(\frac{1}{2}\right)\geq f(1)$

(B) $f\left(\frac{1}{3}\right)\leq f\left(\frac{2}{3}\right)$

(C) $f'(2)\leq 0$

(D) $\dfrac{f'(3)}{f(3)}\geq\dfrac{f'(2)}{f(2)}$

21. If $f(x)=\begin{vmatrix}\cos(2x) & \cos(2x) & \sin(2x)\\ -\cos x & \cos x & -\sin x\\ \sin x & \sin x & \cos x\end{vmatrix}$, then **[JEE (Adv)-2017 (Paper-2)]**

(A) $f(x)$ attains its maximum at $x = 0$

(B) $f(x)$ attains its minimum at $x = 0$

(C) $f'(x) = 0$ at more than three points in $(-\pi, \pi)$

(D) $f'(x) = 0$ at exactly three points in $(-\pi, \pi)$

22. If $f:\mathbb{R}\to\mathbb{R}$ is a differentiable function such that $f'(x) > 2f(x)$ for all $x\in\mathbb{R}$, and $f(0) = 1$, then

[JEE (Adv)-2017 (Paper-2)]

(A) $f(x)$ is increasing in $(0, \infty)$

(B) $f'(x) < e^{2x}$ in $(0, \infty)$

(C) $f(x) > e^{2x}$ in $(0, \infty)$

(D) $f(x)$ is decreasing in $(0, \infty)$

23. For every twice differentiable function $f : R \to [-2, 2]$ with $(f(0))^2 + (f'(0))^2 = 85$, which of the following statement(s) is (are) TRUE? **[JEE (Adv)-2018 (Paper-1)]**

(A) There exist $r, s\in\mathbb{R}$, where $r < s$, such that f is one-one on the open interval (r, s)

(B) There exists $x_0 \in (-4, 0)$ such that $|f'(x_0)| \leq 1$

(C) $\lim\limits_{x\to\infty} f(x)=1$

(D) There exists $\alpha \in (-4, 4)$ such that $f(\alpha) + f''(\alpha) = 0$ and $f'(\alpha) \neq 0$

24. Let $f: \mathbb{R} \to \mathbb{R}$ be given by $f(x) = \begin{cases} x^5 + 5x^4 + 10x^3 + 10x^2 + 3x + 1, & x < 0; \\ x^2 - x + 1, & 0 \le x < 1; \\ \frac{2}{3}x^3 - 4x^2 + 7x - \frac{8}{3}, & 1 \le x < 3; \\ (x-2)\log_e(x-2) - x + \frac{10}{3}, & x \ge 3. \end{cases}$

Then which of the following options is/are correct? **[JEE (Adv)-2019 (Paper-1)]**

(A) f' has a local maximum at $x = 1$
(B) f' is NOT differentiable at $x = 1$
(C) f is onto
(D) f is increasing on $(-\infty, 0)$

25. Let $f: \mathbb{R} \to \mathbb{R}$ be given by $f(x) = (x - 1)(x - 2)(x - 5)$. Define $F(x) = \int_0^x f(t)\,dt,\ x > 0.$ Then which of the following options is/are correct? **[JEE (Adv)-2019 (Paper-2)]**

(A) F has a local maximum at $x = 2$
(B) F has a local minimum at $x = 1$
(C) $F(x) \ne 0$ for all $x \in (0, 5)$
(D) F has two local maxima and one local minimum in $(0, \infty)$

26. Let $f(x) = \frac{\sin \pi x}{x^2}$, $x > 0$. Let $x_1 < x_2 < x_3 < \ldots < x_n < \ldots$ be all the points of local maximum of f and $y_1 < y_2 < y_3 < \ldots < y_n < \ldots$ be all the points of local minimum of f. Then which of the following options is/are correct? **[JEE (Adv)-2019 (Paper-2)]**

(A) $x_1 < y_1$
(B) $x_{n+1} - x_n > 2$ for every n
(C) $|x_n - y_n| > 1$ for every n
(D) $x_n \in \left(2n, 2n + \frac{1}{2}\right)$ for every n

27. Let b be a nonzero real number. Suppose $f: \mathbb{R} \to \mathbb{R}$ is a differentiable function such that $f(0) = 1$. If the derivative f' of f satisfies the equation

$$f'(x) = \frac{f(x)}{b^2 + x^2}$$

for all $x \in \mathbb{R}$, then which of the following statements is/are TRUE? **[JEE (Adv)-2020 (Paper-2)]**

(A) If $b > 0$, then f is an increasing function
(B) If $b < 0$, then f is a decreasing function
(C) $f(x)f(-x) = 1$ for all $x \in \mathbb{R}$
(D) $f(x) - f(-x) = 0$ for all $x \in \mathbb{R}$

28. Let $f: \mathbb{R} \to \mathbb{R}$ be defined by

$$f(x) = \frac{x^2 - 3x - 6}{x^2 + 2x + 4}$$

Then which of the following statements is (are) **TRUE**? **[JEE (Adv)-2021 (Paper-1)]**

(A) f is decreasing in the interval $(-2, -1)$
(B) f is increasing in the interval $(1, 2)$
(C) f is onto
(D) Range of f is $\left[-\frac{3}{2}, 2\right]$

29. Let $|M|$ denote the determinant of a square matrix M. Let $g:\left[0,\frac{\pi}{2}\right]\to\mathbb{R}$ be the function defined by

$$g(\theta)=\sqrt{f(\theta)-1}+\sqrt{f\left(\frac{\pi}{2}-\theta\right)-1}$$ where

$$f(\theta)=\frac{1}{2}\begin{vmatrix}1 & \sin\theta & 1\\ -\sin\theta & 1 & \sin\theta\\ -1 & -\sin\theta & 1\end{vmatrix}+\begin{vmatrix}\sin\pi & \cos\left(\theta+\frac{\pi}{4}\right) & \tan\left(\theta-\frac{\pi}{4}\right)\\ \sin\left(\theta-\frac{\pi}{4}\right) & -\cos\frac{\pi}{2} & \log_e\left(\frac{4}{\pi}\right)\\ \cot\left(\theta+\frac{\pi}{4}\right) & \log_e\left(\frac{\pi}{4}\right) & \tan\pi\end{vmatrix}.$$

Let p(x) be a quadratic polynomial whose roots are the maximum and minimum values of the function $g(\theta)$, and $p(2)=2-\sqrt{2}$. Then, which of the following is/are TRUE ? **[JEE (Adv)-2022 (Paper-1)]**

(A) $p\left(\frac{3+\sqrt{2}}{4}\right)<0$

(B) $p\left(\frac{1+3\sqrt{2}}{4}\right)>0$

(C) $p\left(\frac{5\sqrt{2}-1}{4}\right)>0$

(D) $p\left(\frac{5-\sqrt{2}}{4}\right)<0$

Linked Comprehension Type Questions

Paragraph for Q.Nos. 30 to 32

If a continuous function f defined on the real line **R**, assumes positive and negative values in **R** then the equation $f(x) = 0$ has a root in **R**. For example, if it is known that a continuous function f on **R** is positive at some point and its minimum value is negative then the equation $f(x) = 0$ has a root in **R**.

Consider $f(x) = ke^x - x$ for all real x where k is a real constant. **[IIT-JEE-2007 (Paper-2)]**

Choose the correct answer :

30. The line $y = x$ meets $y = ke^x$ for $k \le 0$ at

(A) No point

(B) One point

(C) Two points

(D) More than two points

31. The positive value of k for which $ke^x - x = 0$ has only one root is

(A) $\frac{1}{e}$

(B) 1

(C) e

(D) $\log_e 2$

32. For $k > 0$, the set of all values of k for which $ke^x - x = 0$ has two distinct roots is

(A) $\left(0,\frac{1}{e}\right)$

(B) $\left(\frac{1}{e},1\right)$

(C) $\left(\frac{1}{e},\infty\right)$

(D) $(0, 1)$

Paragraph for Q.Nos. 33 to 35

Consider the function $f : (-\infty, \infty) \to (-\infty, \infty)$ defined by $f(x) = \dfrac{x^2 - ax + 1}{x^2 + ax + 1}$, $0 < a < 2$. **[IIT-JEE-2008 (Paper-2)]**

Choose the correct answer :

33. Which of the following is true?

(A) $(2 + a)^2 f''(1) + (2 - a)^2 f''(-1) = 0$

(B) $(2 - a)^2 f''(1) - (2 + a)^2 f''(-1) = 0$

(C) $f'(1) f'(-1) = (2 - a)^2$

(D) $f'(1) f'(-1) = -(2 + a)^2$

34. Which of the following is true?

(A) $f(x)$ is decreasing on $(-1, 1)$ and has a local minimum at $x = 1$

(B) $f(x)$ is increasing on $(-1, 1)$ and has a local maximum at $x = 1$

(C) $f(x)$ is increasing on $(-1, 1)$ but has neither a local maximum nor a local minimum at $x = 1$

(D) $f(x)$ is decreasing on $(-1, 1)$ but has neither a local maximum nor a local minimum at $x = 1$

35. Let $g(x) = \displaystyle\int_0^{e^x} \frac{f'(t)}{1+t^2}\,dt$. Which of the following is true?

(A) $g'(x)$ is positive on $(-\infty, 0)$ and negative on $(0, \infty)$

(B) $g'(x)$ is negative on $(-\infty, 0)$ and positive on $(0, \infty)$

(C) $g'(x)$ changes sign on both $(-\infty, 0)$ and $(0, \infty)$

(D) $g'(x)$ does not change sign on $(-\infty, \infty)$

Paragraph for Q.Nos. 36 to 38

Consider the polynomial $f(x) = 1 + 2x + 3x^2 + 4x^3$. Let s be the sum of all distinct real roots of $f(x) = 0$ and let $t = |s|$. **[IIT-JEE-2010 (Paper-2)]**

Choose the correct answer :

36. The real number s lies in the interval

(A) $\left(-\frac{1}{4}, 0\right)$

(B) $\left(-11, -\frac{3}{4}\right)$

(C) $\left(-\frac{3}{4}, -\frac{1}{2}\right)$

(D) $\left(0, \frac{1}{4}\right)$

37. The area bounded by the curve $y = f(x)$ and the lines $x = 0$, $y = 0$ and $x = t$, lies in the interval

(A) $\left(\frac{3}{4}, 3\right)$

(B) $\left(\frac{21}{64}, \frac{11}{16}\right)$

(C) $(9, 10)$

(D) $\left(0, \frac{21}{64}\right)$

38. The function $f'(x)$ is

(A) Increasing in $\left(-t, -\frac{1}{4}\right)$ and decreasing in $\left(-\frac{1}{4}, t\right)$

(B) Decreasing in $\left(-t, -\frac{1}{4}\right)$ and increasing in $\left(-\frac{1}{4}, t\right)$

(C) Increasing in $(-t, t)$

(D) Decreasing in $(-t, t)$

Paragraph for Q.Nos. 39 and 40

Let $f(x) = (1 - x)^2 \sin^2 x + x^2$ for all $x \in \mathbb{R}$, and let $g(x) = \int_1^x \left(\frac{2(t-1)}{t+1} - \ln t\right) f(t)dt$ for all $x \in (1, \infty)$.

[IIT-JEE-2012 (Paper-2)]

Choose the correct answer :

39. Which of the following is true?

(A) g is increasing on $(1, \infty)$

(B) g is decreasing on $(1, \infty)$

(C) g is increasing on $(1, 2)$ and decreasing on $(2, \infty)$

(D) g is decreasing on $(1, 2)$ and increasing on $(2, \infty)$

40. Consider the statements

P : There exists some $x \in \mathbb{R}$ such that $f(x) + 2x = 2(1 + x^2)$.

Q : There exists some $x \in \mathbb{R}$ such that $2f(x) + 1 = 2x(1 + x)$.

Then

(A) Both **P** and **Q** are true

(B) **P** is true and **Q** is false

(C) **P** is false and **Q** is true

(D) Both **P** and **Q** are false

Paragraph for Q.Nos. 41 and 42

Let $f : [0, 1] \to \mathbb{R}$ (the set of all real numbers) be a function. Suppose the function f is twice differentiable, $f(0) = f(1) = 0$ and satisfies $f''(x) - 2f'(x) + f(x) \geq e^x$, $x \in [0, 1]$. **[JEE (Adv)-2013 (Paper-2)]**

Choose the correct answer :

41. Which of the following is true for $0 < x < 1$?

(A) $0 < f(x) < \infty$

(B) $-\frac{1}{2} < f(x) < \frac{1}{2}$

(C) $-\frac{1}{4} < f(x) < 1$

(D) $-\infty < f(x) < 0$

42. If the function $e^{-x} f(x)$ assumes its minimum in the interval $[0, 1]$ at $x = \frac{1}{4}$, which of the following is true?

(A) $f'(x) < f(x), \frac{1}{4} < x < \frac{3}{4}$

(B) $f'(x) > f(x), 0 < x < \frac{1}{4}$

(C) $f'(x) < f(x), 0 < x < \frac{1}{4}$

(D) $f'(x) < f(x), \frac{3}{4} < x < 1$

Paragraph for Q.No. 43

Let $\psi_1 : [0, \infty) \to \mathbb{R}$, $\psi_2 : [0, \infty) \to \mathbb{R}$, $f : [0, \infty) \to \mathbb{R}$ and $g : [0, \infty) \to \mathbb{R}$ be functions such that $f(0) = g(0) = 0$,

$$\psi_1(x) = e^{-x} + x, x \geq 0,$$

$$\psi_2(x) = x^2 - 2x - 2e^{-x} + 2, x \geq 0,$$

$$f(x) = \int_{-x}^{x} \left(|t| - t^2\right) e^{-t^2} dt, x > 0$$

and $g(x) = \int_0^{x^2} \sqrt{t}\, e^{-t}\, dt, x > 0$.

Choose the correct answer :

43. Which of the following statements is TRUE? **[JEE (Adv)-2021 (Paper-2)]**

(A) $f\left(\sqrt{\ln 3}\right) + g\left(\sqrt{\ln 3}\right) = \frac{1}{3}$

(B) For every $x > 1$, there exists an $\alpha \in (1, x)$ such that $\Psi_1(x) = 1 + \alpha x$

(C) For every $x > 0$, there exists a $\beta \in (0, x)$ such that $\Psi_2(x) = 2x\,(\Psi_1(\beta) - 1)$

(D) f is an increasing function on the interval $\left[0, \frac{3}{2}\right]$

Matrix-Match / Matrix Based Type Questions

44. In the following $[x]$ denotes the greatest integer less than or equal to x.

Match the functions in **Column-I** with the properties in **Column-II** and indicate your answer by darkening the appropriate bubbles in the 4 × 4 matrix given in the ORS. **[IIT-JEE-2007 (Paper-1)]**

Column-I	Column-II
(A) $x\|x\|$	(p) continuous in (–1, 1)
(B) $\sqrt{\|x\|}$	(q) differentiable in (–1, 1)
(C) $x + \|x\|$	(r) strictly increasing in (–1, 1)
(D) $\|x-1\| + \|x+1\|$	(s) not differentiable at least at one point in (–1, 1)

Answer Q.45, Q.46 and Q.47 by appropriately matching the information given in the three columns of the following table.

Let $f(x) = x + \log_e x - x\log_e x, x \in (0, \infty)$

- Column 1 contains information about zeros of $f(x), f'(x)$ and $f''(x)$.
- Column 2 contains information about the limiting behaviour of $f(x), f'(x)$ and $f''(x)$ at infinity.
- Column 3 contains information about increasing/decreasing nature of $f(x)$ and $f'(x)$.

Column 1	Column 2	Column 3
(I) $f(x) = 0$ for some $x \in (1, e^2)$	(i) $\lim\limits_{x\to\infty} f(x) = 0$	(P) f is increasing in (0, 1)
(II) $f'(x) = 0$ for some $x \in (1, e)$	(ii) $\lim\limits_{x\to\infty} f(x) = -\infty$	(Q) f is decreasing in (e, e^2)
(III) $f'(x) = 0$ for some $x \in (0, 1)$	(iii) $\lim\limits_{x\to\infty} f'(x) = -\infty$	(R) f' is increasing in (0, 1)
(IV) $f''(x) = 0$ for some $x \in (1, e)$	(iv) $\lim\limits_{x\to\infty} f''(x) = 0$	(S) f' is decreasing in (e, e^2)

45. Which of the following options is the only correct combination? **[JEE (Adv)-2017 (Paper-1)]**

(A) (I) (ii) (R) (B) (IV) (i) (S)

(C) (III) (iv) (P) (D) (II) (iii) (S)

46. Which of the following options is the only correct combination? **[JEE (Adv)-2017 (Paper-1)]**

(A) (I) (i) (P) (B) (II) (ii) (Q)

(C) (III) (iii) (R) (D) (IV) (iv) (S)

47. Which of the following options is the only incorrect combination? **[JEE (Adv)-2017 (Paper-1)]**

(A) (II) (iii) (P) (B) (I) (iii) (P)

(C) (III) (i) (R) (D) (II) (iv) (Q)

Integer / Numerical Value Type Questions

48. The maximum value of the function $f(x) = 2x^3 - 15x^2 + 36x - 48$ on the set $A = \{x \mid x^2 + 20 \le 9x\}$ is

[IIT-JEE-2009 (Paper-2)]

49. Let f be a function defined on $\boldsymbol{R}$ (the set of all real numbers) such that $f'(x) = 2010(x - 2009)(x - 2010)^2 (x - 2011)^3(x - 2012)^4$, for all $x \in \boldsymbol{R}$. If g is a function defined on $\boldsymbol{R}$ with values in the interval $(0, \infty)$ such that $f(x) = \ln(g(x))$, for all $x \in \boldsymbol{R}$, then the number of points in $\boldsymbol{R}$ at which g has a local maximum is

[IIT-JEE-2010 (Paper-2)]

50. Let $f : \mathbb{R} \to \mathbb{R}$ be defined as $f(x) = |x| + |x^2 - 1|$. The total number of points at which f attains either a local maximum or a local minimum is **[IIT-JEE-2012 (Paper-1)]**

51. Let $p(x)$ be a real polynomial of least degree which has a local maximum at $x = 1$ and a local minimum at $x = 3$. If $p(1) = 6$ and $p(3) = 2$, then $p'(0)$ is **[IIT-JEE-2012 (Paper-1)]**

52. The slope of the tangent to the curve $(y - x^5)^2 = x(1 + x^2)^2$ at the point $(1, 3)$ is **[JEE (Adv)-2014 (Paper-1)]**

53. A cylindrical container is to be made from certain solid material with the following constraints: It has a fixed inner volume of V mm^3, has a 2 mm thick solid wall and is open at the top. The bottom of the container is a solid circular disc of thickness 2 mm and is of radius equal to the outer radius of the container.

If the volume of the material used to make the container is minimum when the inner radius of the container is 10 mm, then the value of $\dfrac{V}{250\pi}$ is **[JEE (Adv)-2015 (Paper-1)]**

54. The total number of distinct $x \in [0, 1]$ for which $\int_0^x \frac{t^2}{1+t^4}dt = 2x - 1$ is **[JEE (Adv)-2016 (Paper-1)]**

55. For a polynomial $g(x)$ with real coefficients, let m_g denote the number of distinct real roots of $g(x)$. Suppose S is the set of polynomials with real coefficients defined by

$S = \{(x^2 - 1)^2(a_0 + a_1x + a_2x^2 + a_3x^3) : a_0, a_1, a_2, a_3 \in \mathbb{R}\}$

For a polynomial f, let f' and f'' denote its first and second order derivatives, respectively. Then the minimum possible value of $(m_{f'} + m_{f''})$, where $f \in S$, is _____ **[JEE (Adv)-2020 (Paper-1)]**

56. Let the function $f:(0, \pi) \to \mathbb{R}$ be defined by

$$f(\theta) = (\sin\theta + \cos\theta)^2 + (\sin\theta - \cos\theta)^4,$$

Suppose the function f has a local minimum at θ precisely when $\theta \in \{\lambda_1\pi, ..., \lambda_r\pi\}$, where $0 < \lambda_1 < .... \lambda_r < 1$. Then the value of $\lambda_1 + ... + \lambda_r$ is _______ **[JEE (Adv)-2020 (Paper-2)]**

Question Stem for Question Nos. 57 and 58

Let $f_1 : (0, \infty) \to R$ and $f_2 : (0, \infty) \to R$ be defined by $f_1(x) = \int_0^x \prod_{j=1}^{21} (t-j)^j \, dt, x > 0$ and $f_2(x) = 98(x-1)^{50} - 600(x-1)^{49} + 2450$, $x > 0$, where, for any positive integer n and real number $a_1, a_2, ...a_n$, $\prod_{i=1}^{n} a_i$ denotes the product of $a_1, a_2, ..a_n$. Let m_i and n_i, respectively, denote the number of points of local minima and the number of points of local maxima of function f_i, $i = 1, 2$, in the interval $(0, \infty)$. **[JEE (Adv)-2021 (Paper-2)]**

57. The value of $2m_1 + 3n_1 + m_1n_1$ is _____.

58. The value of $6m_2 + 4n_2 + 8m_2n_2$ is _____.

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